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Data Engineer

PROFESSIONAL SUMMARY

Data Engineer with 4 years of experience building scalable ETL and big data pipelines using the Hadoop ecosystem, IBM DataStage, and AWS.

- Proficient in tools like Hive, Pig, Spark (PySpark and Streaming), Kafka, Sqoop, Flume, and Oozie.
- Created large-scale ingestion and transformation frameworks with Spark, Hive, and DataStage, managing various data formats such as JSON, XML, and Parquet.
- Have a strong background in data modeling (SCD, Star schema), data quality frameworks, and handling errors.
- Worked on Agile (Scrum and Kanban) projects, collaborating closely with vendors, BAs, and DevOps teams.
- Have hands-on experience with AWS EMR, EC2, S3, version control tools (Git, Bitbucket), and job schedulers (Control-M, Autosys, Oozie).

SKILL MATRIX

Big Data (Hadoop Ecosystem) Tools	Hive, Pig, Oozie, Flume, PySpark, Hadoop HDFS, Apache Storm, Apache Spark, Sqoop, Apache Avro, Apache Kafka, Apache HBase, Zookeeper, nifi
ETL Tools	IBM Infosphere DataStage, Informatica
Cloud Computing Services	Elastic Map Reduce, EC2, S3, Athena, Microsoft Azure, Snowflake
Web Development Tools	PyCharm, VS Code, Jupyter, Cloudera Data Science Workbench
Build Tools	Control-M, Oozie, Autosys, Talend
Databases	Oracle, SQL Server, Flat Files, Teradata RDBMS
No SQL Databases	Cassandra, Mango DB, HBase
Development Methodologies	Agile Scrum, Kanban Agile, Waterfall
Languages	XML, JSON, SAS, Shell, HTML, Java, SQL, PL/SQL, SAS Enterprise Guide
Reporting Tools	MS Office (Word/Excel/PowerPoint/ Visio/Outlook), Crystal Reports XI, SSRS, Cognos.
Operating Systems	Windows, UNIX, LINUX

PROFESSIONAL EXPERIENCE

Fannie Mae, Washington, DC

Apr 2023 to Present

Data Engineer

Responsibilities:

- Analyzed source system structures and mapped over 500 data elements to target systems. This ensured complete compliance with business needs for data migration and transformation.
- Developed and deployed Spark applications using PySpark and Spark SQL. These applications processed and transformed more than 4TB of data each week, revealing important insights into customer usage patterns and leading to product improvements.
- Managed a Cloudera-based Hadoop infrastructure and supported 24/7 production systems across Hadoop, Spark, NiFi, and Impala. Achieved 99.95% uptime through proactive monitoring and tuning.
- Applied machine learning algorithms, such as Random Forest, Decision Tree, and Linear Regression, on formatted data for predictive modeling.
- This enabled better customer segmentation and more targeted marketing. Used Scala and Play Framework to build APIs and real-time data connectors. Handled over 100,000 events per minute with Kafka and Flink for real-time ingestion and processing pipelines.
- Created and executed Python scripts to parse and transform large volumes of JSON and log data, making it easier

to ingest into downstream databases and analytics tools.

- Automated CI/CD pipelines with Jenkins, cutting manual deployment effort by 70% and speeding up release cycles for Spark and Hive jobs. Wrote Scala-based validation logic to enforce data quality rules on log data streams. This prevented bad data from spreading and improved data integrity by 95%. Conducted data profiling and cleansing across Hadoop, Hive, and Impala layers using PySpark.
- This resulted in a 30% reduction in data-related incidents after deployment. Performed gap analysis and ETL impact assessments for new business requirements. Successfully improved over 50 existing jobs without causing regression issues.
- Designed and executed unit tests using mock data in lower environments. This helped find bugs early and cut the production rollback rate by 40%. Built Sqoop and PySpark-based ingestion pipelines for both historical and incremental data, integrating them into Control-M workflows for scheduled execution. Authored and updated detailed design documentation for each production release.
- Maintained issue trackers to log, troubleshoot, and analyze ETL failures. Tuned long-running Spark jobs by optimizing partitions, memory settings, and SQL queries, reducing processing time on some jobs from 2 hours to under 30 minutes. Participated in code reviews, framework design, and deployment planning.
- Coordinated with Hadoop Admins, ETL Developers, and SAS teams to ensure production readiness. Handled Remedy/ServiceNow tickets for production issues, job reruns, environment deployments, and scheduled FTP/ETL triggers on a daily, weekly, and monthly basis.
- Created reusable UNIX shell scripts for validation, archiving, and control flow logic in Hadoop workflows, cutting scripting time by 50% across similar projects.

Abbvie Inc. – Hyd, India

Sept 2021 to Aug 2022

Data Engineer Responsibilities:

- Hands-on major components in Hadoop Echo Systems like Spark, HDFS, HIVE, HBase, Zookeeper, Sqoop, and Oozie.
- Developing Sqoop jobs to ingest data from various systems of records into Enterprise Data Lake. Development of Spark jobs in PySpark and SparkSQL to run on top of hive tables and create transformed data sets for downstream consumption.
- Working with business analysts to convert functional requirements into technical requirements and build appropriate data pipelines.
- Ingesting data from various source systems like Oracle, SQL Server, Flat files, JSONs.
- Conducting exploratory data analysis in Jupyter notebooks using Python libraries and sharing the data analysis.
- Performance tuning spark and hive jobs by reading execution plans, DAGs, and Yarn logs.
- Creating generic shell scripts to submit Hadoop and spark jobs on EMR and on-prem edge node.
- Worked on migrating on-prem Hadoop cluster data and data pipelines to AWS Cloud and migration project which included migrating web methods code to Informatica cloud.
- Writing Complex SparkSQL codes to clean, join, transform, and aggregate the datasets and publish them for Power BI team to produce operational scorecards.
- Writing custom python modules for reusable python code.
- Working on designing the MapReduce and Yarn flow and writing MapReduce scripts, performance tuning and debugging.
- Created multiple proof-of-concepts using PySpark and deployed them on the Yarn cluster, comparing Spark's performance to that of Hive and SQL/Teradata.
- Worked on setting up AWS DMS and SNS for data transfer and replication and used SQL on the new AWS Databases like RedShift and Relation Data Services.
- Integrated Teradata Warehouse into EMR cluster. Developed BTEQ scripts to load data from Teradata Staging area to Teradata data mart. Handled Error & tuned performance in Teradata queries and utilities.
- Set-up databases in GCP using RDS, storage using S3 bucket and configuring instance backups to S3 bucket. Prototype CI/CD system with GitLab on GKE utilizing Kubernetes and Docker for the runtime environment for the CI/CD systems to build and test and deploy.
- Solving appropriate partition, bucketing schemes, and making sure correct load policies are employed so data can be stored as per requirements.
- Generates ETL scripts to transform, flatten, and enrich the data from source to target using AWS Glue and created

event driven ETL pipelines with AWS Glue.

- Created external tables with partitions using Hive, AWS Athena, and Redshift.
- Created and deployed Lambda functions in AWS using pre-built AWS Lambda Libraries, as well as Lambda functions in Scala using custom libraries.
- Optimizing and tuning the Redshift environment, enabling queries to perform up to 100x faster for Tableau and SAS Visual Analytics.
- Creating Oozie workflows, Coordinators, and scheduling handshake jobs in Control-M.
- Created Session Beans and controller Servlets for handling HTTP requests from Talend.
- Working with production support teams and administration teams to ensure correct access controls are set up on each database.
- PL/SQL is used for writing triggers to perform specific actions when data is inserted, updated, or deleted.
- During database operations, failures and exceptions may occur. These are gracefully handled using PL/SQL's exception handling features. Working with governance teams to ensure metadata management, data lineage and technical metadata are correctly updated for each data asset.
- Working with the master-feature branch model and committing the code with appropriate comments.
- Attending sprint planning, agile ceremonies, and demoting the work products on a bi-weekly basis. Documenting data flow diagrams and technical logic in Confluence.

Environment: Cloudera Hadoop distribution, AWS EMR, Glue, S3, Athena, Hive, Impala, Lambda, Scala, PySpark, Spark SQL, Oracle 11g/12c, Teradata, Jira, Bitbucket, Power BI, Control-M- M, Talend, CI/CD pipelines, GKE, Redshift, Informatica, SAS

Azurity – Hyd, India

Mar 2020 to Sept 2021

Data Engineer

Responsibilities:

- Built and maintained over 50 Business Objects reports and executive dashboards. I integrated these with Teradata and Oracle, cutting manual reporting time by 70%.
- Designed and implemented more than five end-to-end BI solutions on Azure. I used Data Factory, Synapse, Databricks, ADLS, U-SQL, and HDInsight. These solutions supported multi-TB datasets.
- Led the migration of the data warehouse from Oracle 9i to 10g.
- Optimized database performance and took advantage of new features like DBMS_SCHEDULER and Data Pump.
- Created NoSQL databases in MongoDB with sharding and replication. This improved query performance by 40% for high-volume applications.
- Developed Python and Spark pipelines for data cleansing and transformation on Azure HDInsight. This increased data processing speeds by 60% compared to MapReduce.
- Built predictive models using Python with pandas, NumPy, and scikit-learn.
- Visualized insights with Plotly, integrating them with SAS and MSSQL for real-time updates. I encoded and decoded JSON data using PySpark.
- Created custom DataFrames and managed unstructured data with HBase and NoSQL ingestion from UNIX systems.
- Automated data ingestion from multiple sources to Azure SQL, DW, and Storage. I used ADF pipelines, T-SQL, and U-SQL, enabling near real-time analytics.
- Developed universes and reports using BO XI R2 and 6.5. I resolved loops and context issues while supporting ad-hoc user queries from more than three business units.
- Have experience with Docker and Kubernetes on GCP.
- Support containerized BI workloads and use Azure Synapse for distributing predictive workloads.

EDUCATION:

MASTERS: Business analytics and information systems

(University Of South Florida Tampa FL)